

Biomolecular Condensates and the Origin of Life

From Flory–Huggins thermodynamics to protocellular compartments — first-principles mathematical modelling of liquid–liquid phase separation in biology

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Abstract

Biomolecular condensates are **membrane-free liquid droplets** inside cells, formed by liquid–liquid phase separation (LLPS) of proteins and nucleic acids. Discovered as P granules in *C. elegans* germ cells (Brangwynne et al., 2009), they have since been found to organise virtually every aspect of cellular function: gene regulation, signal transduction, stress response, and the formation of structures from the nucleolus to synaptic densities.

This notebook develops the mathematical physics of condensates from first principles. We implement the **Flory–Huggins** free energy to derive the phase diagram, solve the **Cahn–Hilliard** partial differential equation for spinodal decomposition, compute the **concentration enhancement** that makes condensates natural chemical reactors, and model the **protocell dynamics** (growth, division, competition) that connect condensates to the origin of life. We conclude with the **liquid → gel → aggregate** transition that links condensate dysfunction to neurodegenerative disease.

Audience: the prose is written for two overlapping audiences. Each section opens with an intuitive explanation that requires no background, then deepens into a quantitative treatment with equations, numerical integration, and visualisations. Biophysicists will recognise the canonical references; readers from other backgrounds should be able to follow the physical narrative.

All code and figures are original and open-source: github.com/mthiel74/biomolecular-condensates (<https://github.com/mthiel74/biomolecular-condensates>).

Roadmap. §2–3 build the equilibrium and dynamical theory (Flory–Huggins phase diagrams, Cahn–Hilliard spinodal decomposition and coarsening). §4–6 turn condensates into chemistry: concentration enhancement, an RNA-world replication window, and protocell growth, division and proto-Darwinian selection. §7 covers the liquid→gel→aggregate disease transition. §8–10 add the nonlinear-dynamics toolkit (stability, bifurcation, Lyapunov functionals), active/chemically-maintained condensates, and the emergent layer where selection, Turing patterns and wet–dry cycles arise. §11 concludes.

Initialisation — evaluate this cell first. It loads the figure package `condensate_figures.wl`, which is attached to this post: download it into the *same folder* as this notebook (the loader also resolves the repository layout automatically). Every figure below is then generated by the code cell shown just above it.

In[]:=*

```
Module[
  {file = "condensate_figures.wl",
   nbDir = Quiet@NotebookDirectory[], cwd = Quiet@Directory[], candS, found},
  candS = Select[{
    If[StringQ[nbDir], FileNameJoin[{nbDir, file}], Null],
    If[StringQ[nbDir],
      FileNameJoin[{ParentDirectory[nbDir], "wolfram", file}], Null],
    If[StringQ[cwd], FileNameJoin[{cwd, file}], Null],

    "/Users/thiel/GitHub/biomolecular-condensates/wolfram/condensate_figures.wl"
  ], StringQ];
  found = SelectFirst[candS, FileExistsQ];
  If[StringQ[found], Get[found],
  Print[
    "condensate_figures.wl not found - download it from this post and place
    it in the same folder as this notebook."]]]
```

In[]:=*

```
figParamSpace[]
```

Out[]=*

```
–Graphics–
```

The phase behaviour of biomolecular condensates, generated live here: (left) the interaction–chain-length plane that decides whether a mixture demixes at all, and (right) how strongly a condensate accelerates the reactions inside it. Beneath it all runs a single idea — the same Flory–Huggins physics that unmixes oil from water also organises the living cell, and may have lit the first spark of life.

1. The membrane-free organelle

For over a century, cell biology was built on the assumption that the cell's internal organisation depends on **membranes**: lipid bilayers that enclose the nucleus, the mitochondria, the endoplasmic reticulum. Compartmentalisation by membranes was the organising principle of eukaryotic life.

Then, in 2009, **Brangwynne, Eckmann, Courson, Rybarska, Hoege, Gharakhani, Jülicher, and Hyman** showed that P granules — RNA–protein bodies in the germ cells of the nematode *C. elegans* — are not membrane-bound organelles at all. They are **liquid droplets**. They flow, they fuse, they drip, they dissolve and re-form. They are phase-separated condensates of RNA and proteins, held together not by a lipid membrane but by the same physics that separates oil from water.

This discovery launched a revolution. Within a decade, dozens of cellular structures were reclassified as condensates: stress granules, nucleoli, Cajal bodies, PML bodies, centrosomes, synaptic densities, heterochromatin domains, transcriptional hubs. Protein science has been transformed in parallel — the 2024 Nobel Prize in Chemistry went to David Baker for computational protein design and jointly to Demis Hassabis and John Jumper for protein-structure prediction — and the conden-

sate revolution adds a complementary lesson: that proteins are not only rigid machines but can phase-separate into functional liquids.

The physics. The formation of a condensate is a thermodynamic phase transition. Below a critical temperature (or above a critical interaction strength), a homogeneous solution of macromolecules becomes thermodynamically unstable and spontaneously separates into a dilute phase and a dense phase. The dense phase is the condensate. The transition is governed by a competition between the entropy of mixing (which favours homogeneity) and the enthalpic interactions between molecules (which favour like-with-like aggregation). The mathematical framework for this competition is **Flory–Huggins theory**, which we derive in the next section.

The origin-of-life connection. Perhaps the most provocative implication of the condensate revolution is its connection to the **origin of life**. Aleksandr Oparin proposed (in 1924, and in molecular detail in his 1936 book) that life began inside **coacervate droplets** — spontaneously forming colloidal structures that concentrate organic molecules from the dilute primordial ocean. Oparin's hypothesis was dismissed for decades because the known coacervates were too disordered to sustain biochemistry. But modern condensates, with their exquisite sequence-dependent phase behaviour and their ability to concentrate specific clients by factors of up to a thousand, revive Oparin's vision in molecular detail. We show in §4 that condensate-assisted chemistry solves the **concentration problem** that has bedevilled prebiotic chemistry: how do you get reactions going in a picomolar ocean?

2. Thermodynamics of phase separation

2.1 The Flory–Huggins free energy

The lattice model. Consider a lattice of N_t sites, each occupied by either a monomer of species 1 (the polymer, chain length N_1) or a monomer of species 2 (the solvent, with $N_2 = 1$). Let ϕ be the volume fraction of species 1. The Flory–Huggins free energy of mixing per lattice site is:

$$\frac{f}{kT} = \frac{\phi \ln \phi}{N_1} + \frac{(1 - \phi) \ln(1 - \phi)}{N_2} + \chi \phi(1 - \phi)$$

The first two terms are the **entropy of mixing** — always negative (favour mixing) and scaled by the inverse chain lengths $1/N_1$ and $1/N_2$. Longer chains have less translational entropy per monomer, so polymers phase-separate more readily than small molecules. The third term is the **enthalpy of mixing**, parameterised by the Flory interaction parameter χ . For many condensate-forming proteins, $\chi > 0$ (unfavourable mixing) and χ decreases with temperature ($\chi \propto 1/T$), giving an upper critical solution temperature (UCST) phase diagram — though a substantial class of intrinsically disordered proteins (e.g. elastin-like sequences) instead show the opposite, lower critical solution temperature (LCST), behaviour.

```

In[ ]:=
  figFHFreeEnergy[100, 1]

Out[ ]:=
  Legended[Graphics, Placed[LineLegend[{Directive[Thickness[Large],
    >   RGBColor[0., 0., 0.7]],
    >   Directive[Thickness[Large], RGBColor[0, 0, 1]],
    >   Directive[Thickness[Large], RGBColor[0,  $\frac{2}{3}$ , 0]],
    >   Directive[Thickness[Large], RGBColor[ $\frac{2}{3}$ , 0.333333, 0]],
    >   Directive[Thickness[Large], RGBColor[ $\frac{2}{3}$ , 0, 0]]},
    >   { $\bar{\chi}$  = 0.182,  $\bar{\chi}$  = 0.303,  $\bar{\chi}$  = 0.605,  $\bar{\chi}$  = 0.907,  $\bar{\chi}$  = 1.510},
    >   LabelStyle -> Directive[GrayLevel[1], 11],
    >   LegendLayout -> {Column, {Automatic, 5}}, LegendMarkerSize -> 22,
    >   Background -> Directive[GrayLevel[0, 0.55]]], Scaled[{0.52, 0.74}],
    >   Identity]]

```

Figure 2.1 Flory-Huggins free energy density for a polymer-solvent system ($N_1 = 100$, $N_2 = 1$) at several values of χ . Below the critical χ , the free energy has a single minimum (stable mixed state). Above it, a double well appears: the system phase-separates into dilute (ϕ_l) and dense (ϕ_d) phases at the two minima.

Evaluate the cell below to reproduce Figure 2.1 — the whole of Flory-Huggins is two lines of Wolfram Language:

```

In[ ]:=
  fFH[phi_, n1_, n2_, chi_] := phi Log[phi] / n1 + (1 - phi) Log[1 - phi] / n2 + chi phi (1 - phi);
  Plot[Evaluate@Table[fFH[phi, 100, 1, chi], {chi, {0.4, 0.605, 0.9, 1.5}}],
    {phi, 0.001, 0.999},
    PlotRange -> {-0.04, 0.04}, Frame -> True, FrameLabel -> {"phi", "f/kT"},
    PlotLegends -> {"chi=0.4", "chi=0.605 (critical)", "chi=0.9", "chi=1.5"}]

Out[ ]:=
  Legended[Graphics, Placed[LineLegend[{Directive[Opacity[1.],
    >   RGBColor[0.24, 0.6, 0.8], AbsoluteThickness[2]],
    >   Directive[Opacity[1.], RGBColor[0.95, 0.627, 0.1425],
    >   AbsoluteThickness[2]], Directive[Opacity[1.],
    >   RGBColor[0.455, 0.7, 0.21], AbsoluteThickness[2]],
    >   Directive[Opacity[1.], RGBColor[0.922526, 0.385626, 0.209179],
    >   AbsoluteThickness[2]]},
    >   { $\bar{\chi}$  = 0.4,  $\bar{\chi}$  = 0.605 (critical),  $\bar{\chi}$  = 0.9,  $\bar{\chi}$  = 1.5}, LegendMarkers -> None,
    >   LabelStyle -> {}, LegendLayout -> Column], After, Identity]]

```

And the double well opening up as χ crosses the critical value is best felt interactively (drag χ through ≈ 0.6):

In[]:=

```
Manipulate[
  Plot[ $\phi \text{Log}[\phi] / n1 + (1 - \phi) \text{Log}[1 - \phi] + \chi \phi (1 - \phi)$ ,
    { $\phi$ , 0.001, 0.999}, PlotRange -> All, Frame -> True,
    FrameLabel -> {" $\phi$ ", "f/kT"}, ImageSize -> 480],
  {{ $\chi$ , 0.605, " $\chi$ "}, 0.3, 1.6, 0.005},
  {{n1, 100, "chain length N"}, 1, 300, 1}]
```

Out[]:=

```
Manipulate[Plot[ $\frac{\text{Log}[\tilde{\phi}]}{n1} + (1 - \tilde{\phi}) \text{Log}[1 - \tilde{\phi}] + \tilde{\phi} \tilde{\chi} (1 - \tilde{\phi})$ ,
  > { $\tilde{\phi}$ , 0.001, 0.999}, PlotRange -> All, Frame -> True,
  > FrameLabel -> { $\tilde{\phi}$ , f/kT}, ImageSize -> 480],
  > {{ $\tilde{\phi}$ , 0.605,  $\tilde{\phi}$ }, 0.3, 1.6, 0.005},
  > {{n1, 100, chain length N}, 1, 300, 1}]
```

2.2 Spinodal and binodal: the phase diagram

The thermodynamic stability of the mixed state is determined by the curvature of the free energy.

The **spinodal** is the locus of inflection points:

$$\frac{\partial^2 f}{\partial \phi^2} = \frac{1}{N_1 \phi} + \frac{1}{N_2 (1 - \phi)} - 2\chi = 0$$

Inside the spinodal, the free energy has negative curvature ($\partial^2 f / \partial \phi^2 < 0$) and the system is **unstable** to infinitesimal composition fluctuations — this is **spinodal decomposition**. Between the spinodal and the **binodal** (the coexistence curve), the system is **metastable** — it requires a finite-amplitude fluctuation (nucleation) to phase-separate.

The binodal is found by the **common tangent construction**: the two coexisting compositions ϕ_l and ϕ_d satisfy equal chemical potential and equal osmotic pressure:

$$\mu(\phi_L) = \mu(\phi_R), \quad \Pi(\phi_L) = \Pi(\phi_R)$$

The **critical point** is where the spinodal and binodal meet:

$$\phi_c = \frac{\sqrt{N_2}}{\sqrt{N_1} + \sqrt{N_2}}, \quad \chi_c = \frac{\left(\frac{1}{\sqrt{N_1}} + \frac{1}{\sqrt{N_2}} \right)^2}{2}$$

For a long polymer ($N_1 = 100, N_2 = 1$): $\phi_c = 0.091, \chi_c = 0.605$. The critical concentration is low (~9%), meaning that even dilute polymer solutions phase-separate when χ exceeds a modest threshold.

```

In[*]:=
figFHPhaseDiagram[100, 1]

Out[*]=
Legended[Legended[Legended[-Graphics-,
>   Placed[LineLegend[{Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2],
>   RGBColor[0, 0, 1], Thickness[Large]],
>   Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2], RGBColor[0, 0, 1],
>   Thickness[Large]]], {Binodal, None},
>   LegendMarkers -> {{False, Automatic}}, Joined -> {True, True},
>   LabelStyle -> {}, LegendLayout -> Column], After, Identity]],
>   Placed[LineLegend[{Directive[PointSize[ $\frac{1}{90}$ ], AbsoluteThickness[2],
>   RGBColor[1, 0, 0], Dashing[{Small, Small}], Thickness[Large]]],
>   {Spinodal}, LegendMarkers -> {{False, Automatic}}, Joined -> {True},
>   LabelStyle -> {}, LegendLayout -> Column], After, Identity]],
>   Placed[PointLegend[{Directive[AbsoluteThickness[2], GrayLevel[0],
>   PointSize[Large]]], {Critical point},
>   LegendMarkers -> {{False, Automatic}}, Joined -> {False},
>   LabelStyle -> {}, LegendLayout -> Column], After, Identity]]

```

Figure 2.2 Phase diagram in χ - ϕ coordinates. The blue curve is the binodal (coexistence); the red dashed curve is the spinodal (stability limit). Between them: metastable (nucleation and growth). Inside the spinodal: unstable (spinodal decomposition). The black dot is the critical point.

2.3 Chain length and multivalency

A key prediction of Flory–Huggins theory is that **longer polymers phase-separate more readily**. The critical interaction falls toward $1/2$ as chain length grows, with the excess $\chi^c - 1/2$ scaling as $N^{(-1/2)}$: a 1000-mer demixes at $\chi^c \approx 0.53$ versus ≈ 0.87 for a 10-mer, so long chains phase-separate at appreciably weaker interactions. This is biologically significant: intrinsically disordered regions (IDRs) of condensate-forming proteins are typically 50–500 residues, and their effective N is further increased by multivalent interactions (multiple binding sites per molecule, RNA scaffolds, post-translational modifications).

```

In[*]:=
figFHChainLength[{1, 5, 20, 100, 500}]

Out[*]=
Legended[Legended[Legended[Legended[Legended[-Graphics-,
>     Placed[LineLegend[{Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2],
>         Thickness[Large], RGBColor[0.237736, 0.340215, 0.575113]]},
>         {N = 1}, LegendMarkers -> {{False, Automatic}}, Joined -> {True},
>         LabelStyle -> {}, LegendLayout -> Column], After, Identity]],
>     Placed[LineLegend[{Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2],
>         Thickness[Large], RGBColor[0.277947, 0.450097, 0.328156]]},
>         {N = 5}, LegendMarkers -> {{False, Automatic}}, Joined -> {True},
>         LabelStyle -> {}, LegendLayout -> Column], After, Identity]],
>     Placed[LineLegend[{Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2],
>         Thickness[Large], RGBColor[0.624866, 0.673302, 0.264296]]},
>         {N = 20}, LegendMarkers -> {{False, Automatic}}, Joined -> {True},
>         LabelStyle -> {}, LegendLayout -> Column], After, Identity]],
>     Placed[LineLegend[{Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2],
>         Thickness[Large], RGBColor[0.845341, 0.62487, 0.315178]]},
>         {N = 100}, LegendMarkers -> {{False, Automatic}}, Joined -> {True},
>         LabelStyle -> {}, LegendLayout -> Column], After, Identity]],
>     Placed[LineLegend[{Directive[PointSize[ $\frac{1}{72}$ ], AbsoluteThickness[2],
>         Thickness[Large], RGBColor[0.72987, 0.239399, 0.230961]]}, {N = 500},
>         LegendMarkers -> {{False, Automatic}}, Joined -> {True},
>         LabelStyle -> {}, LegendLayout -> Column], After, Identity]]]

```

Figure 2.3 Spinodal curves for different chain lengths. Longer chains produce wider two-phase regions at higher temperatures (lower χ threshold). This is why proteins with long disordered regions are prone to phase separation.

3. Cahn–Hilliard dynamics of droplet formation

3.1 The equation

The Flory–Huggins theory tells us *whether* phase separation occurs, but not *how*. The dynamics of a conserved order parameter (composition must be conserved — matter doesn't appear or disappear) are governed by the **Cahn–Hilliard equation**:

$$\frac{\partial \phi}{\partial t} = M \nabla^2 \left(\frac{\delta F}{\delta \phi} - \kappa \nabla^2 \phi \right)$$

where M is the mobility (diffusion coefficient), $\delta F/\delta\phi$ is the chemical potential from the Flory–Huggins free energy, and κ is the gradient energy coefficient that penalises sharp interfaces (it sets the surface tension γ of the condensate).

Physical content. The Cahn–Hilliard equation says: material flows down chemical potential gradients (as in ordinary diffusion), but the chemical potential now includes a penalty for composition gradients (the κ term). Inside the spinodal region, the bulk chemical potential drives **uphill diffusion** — material flows from dilute to concentrated regions, amplifying any fluctuation. The gradient penalty prevents infinitely sharp interfaces and selects a characteristic wavelength for the initial instability.

3.2 Spinodal decomposition

We solve the 1D Cahn–Hilliard equation using Wolfram's built-in `NDSolve` with the Method of Lines. The Flory–Huggins chemical potential is logarithmically stiff near the pure phases and the strongly-stabilised 1D solver freezes; we therefore use the standard symmetric double-well (Model B) order parameter ψ , started from a uniform state plus a small multi-mode periodic perturbation, and map it to a physical volume fraction $\phi = \phi_0 + A\psi$ for display.

```
mu[ψ_] := -a ψ + b ψ^3; (* double-well Model B *)
sol = NDSolveValue[{
  D[ψ[x, t], t] == M D[
    mu[ψ[x, t]] - κ D[ψ[x, t], x, x], x, x],
  ψ[x, 0] == icf[x], (* smooth periodic perturbation *)
  ψ[0, t] == ψ[L, t], (* periodic BCs *)
  ψ'[0, t] == ψ'[L, t],
  ψ, {x, 0, L}, {t, 0, T},
  Method -> {"MethodOfLines",
    "SpatialDiscretization" -> {"TensorProductGrid",
      "MinPoints" -> 256}}];
```

In[]:=

figCH1DEvolution[]

Out[]:=

–Graphics–

Figure 3.1 Top: space–time density plot of the 1D spinodal decomposition. The initially uniform composition rapidly develops concentration fluctuations that coarsen into well-separated dense (red) and dilute (blue) domains. Bottom: composition profiles at selected times showing the progression from small perturbation to fully phase-separated condensates.

3.3 Coarsening: Lifshitz–Slyozov–Wagner

Once phase separation is complete, the system continues to evolve: large droplets grow at the expense of small ones. This is **Ostwald ripening**, described by the Lifshitz–Slyozov–Wagner (LSW) theory. The critical radius R^* is set by the Gibbs–Thomson equation:

$$c_{\text{eq}}(R) = c_{\infty} \exp\left(\frac{2\gamma V_m}{RkT}\right) \approx c_{\infty} \left(1 + \frac{l_c}{R}\right)$$

Droplets smaller than the critical radius ($R < R^*$) have an equilibrium concentration above the surrounding matrix supersaturation — they lose material and dissolve. Droplets with $R > R^*$ sit below it — they grow. The mean domain size grows as $\langle R \rangle \propto t^{1/3}$, the classic LSW scaling.


```
In[*]:=
figProtoOstwald[]

Out[*]=
-Graphics-
```

Figure 3.2 Ostwald ripening of a polydisperse condensate population (100 droplets). Left: the mean and maximum droplet radius grow with time as large droplets feed on small ones. Right: the number of surviving droplets falls as the smallest dissolve — the hallmark of LSW coarsening ($\langle R \rangle \propto t^{1/3}$).

4. Concentration enhancement and catalysis

4.1 The partition coefficient

When a condensate forms, molecules partition between the dense and dilute phases according to their chemical affinity. The **partition coefficient** K is defined as:

$$K = \frac{C_{\text{in}}}{C_{\text{out}}}$$

For condensate-forming proteins, K is typically 10–100. For RNA molecules that interact strongly with the condensate scaffold, K can reach 100–1000. A molecule at 1 nM bulk concentration with $K = 100$ is at 100 nM inside the condensate. A molecule at 1 nM with $K = 1000$ is at 1 μM . This is the concentration miracle.

4.2 Bimolecular rate enhancement

For a bimolecular reaction $A + B \rightarrow C$ with second-order rate constant k :

$$\text{rate}_{\text{dilute}} = k [A]_{\text{out}} [B]_{\text{out}}$$

$$\text{rate}_{\text{condensate}} = k K_A [A]_{\text{out}} K_B [B]_{\text{out}} = K_A K_B \text{rate}_{\text{dilute}}$$

If both reactants partition equally ($K_A \approx K_B \approx K$), the reaction rate is enhanced by K^2 . With $K = 100$, this is a **10⁴-fold enhancement**. This is the *open-system* (buffered-reservoir) limit, and it is an upper bound: it assumes the surrounding dilute phase is never depleted. As §8.5 shows, imposing mass conservation caps the realised gain at order $K/4$. Even so, the mechanism reframes the concentration problem in prebiotic chemistry: you do not need a concentrated primordial soup if you have phase separation.

```
In[*]:=
figRateEnhancement[]

Out[*]=
Legended[-Graphics-, Placed[LineLegend[{RGBColor[0, 0, 1],
> RGBColor[1, 0, 0]}, {Bimolecular (K^2),
> Termolecular (K^3, illustrative)}, LabelStyle -> {}],
> LegendLayout -> Column], Scaled[{0.3, 0.8}], Identity]]
```

Figure 4.1 Reaction rate enhancement inside condensates as a function of the partition coefficient K , in the open-system limit (mass conservation lowers the realised gain — see §8.5). Bimolecular reactions (blue) scale as K^2 ; the termolecular curve (red, K^3) is shown only to illustrate the scaling, as three-body reactions are kinetically negligible in solution. At the typical condensate $K \approx 100$, the bimolecular bound is 10^4 .

4.3 Prebiotic chemistry phase space

Which combinations of partition coefficient and bulk concentration allow net chemical production?
We map the net replication rate (replication minus degradation) across the K-c parameter space:

```
In[*]:=
figPrebioticSpace[1.0]

Out[*]=
-Graphics-
```

Figure 4.2 Net replication rate as a function of partition coefficient K and bulk substrate concentration. The black contour marks the boundary between net growth (blue) and net decay (red). At low bulk concentrations (the likely prebiotic regime), only condensates with $K > 50$ – 100 support net replication.

5. RNA world inside condensates

5.1 Ribozyme kinetics

The RNA world hypothesis proposes that the first self-replicating molecules were RNA ribozymes — RNA molecules that catalyse their own replication. The kinetics of ribozyme-catalysed replication follow **Michaelis–Menten** saturation:

$$v = \frac{V_{\max} [S]}{K_m + [S]}$$

Inside a condensate, both the enzyme (ribozyme) and the substrate (template RNA) are concentrated by the partition coefficient K . Writing $V_{\max}$ for the dilute-phase maximal rate (itself proportional to enzyme concentration), the in-condensate $V_{\max}$ is raised by the factor K (more enzyme) and the substrate concentration $[S]$ is likewise raised by K (more substrate):

$$v_{\text{condensate}} = \frac{K V_{\max} K [S]}{K_m + K [S]}$$

At low substrate concentrations ($K [S] \ll K_m$), the rate scales as K^2 — the full bimolecular enhancement. At saturation ($K [S] \gg K_m$), the rate scales as K (only the enzyme enhancement matters).

5.2 Replication vs degradation

RNA is chemically unstable. In the prebiotic ocean, hydrolysis degrades RNA on timescales of hours to days. The **origin-of-life window** exists only when the replication rate exceeds the degradation rate:

$$v_{\text{rep}} > v_{\text{deg}} \iff K > K^*$$

where K^* is the critical partition coefficient. Below K^* , degradation wins and no self-replicating system can persist.

```

In[*]:=
figRibozyme[0.5]

Out[*]=
Legended[Graphics, Placed[LineLegend[{Directive[PointSize[ $\frac{1}{120}$ ],
> AbsoluteThickness[2], Thickness[Large], RGBColor[0, 0, 1]],
> Directive[PointSize[ $\frac{1}{120}$ ], AbsoluteThickness[2], Thickness[Large],
> Dashing[{Small, Small}], RGBColor[1, 0, 0]]},
> {Replication rate, Degradation rate},
> LegendMarkers -> {{False, Automatic}, {False, Automatic}},
> Joined -> {True, True}, LabelStyle -> {}, LegendLayout -> Column],
> After, Identity]]

```

Figure 5.1 Replication (blue) versus degradation (red) rate as a function of the partition coefficient K , at fixed bulk substrate $[S] = 1 \mu\text{M}$. Replication is enhanced as K^2 and degradation as K , so above a critical K^* the green-shaded origin-of-life window opens: replication outpaces hydrolysis and a self-replicating system can persist.

5.3 The origin-of-life window

The critical partition coefficient K^* depends on the ribozyme efficiency (V_{max}/K_m), the degradation rate constant, and the substrate level. For plausible prebiotic parameters ($V_{\text{max}} \approx 1 \mu\text{M}/\text{min}$, $K_m \approx 10 \mu\text{M}$, $k_{\text{eo}}^d \approx 0.5/\text{min}$), we find $K^* \approx 10$ at $[S] = 1 \mu\text{M}$, rising toward 50–100 at the lower substrate levels of a dilute prebiotic pool. Modern coacervate droplets routinely achieve $K = 50$ –500, comfortably above this threshold — the concentration-enhancement mechanism is not marginal.

This is a quantitative version of Oparin's coacervate hypothesis: phase separation concentrates reactants enough to bootstrap biochemistry from a dilute ocean — though, as §8.5 shows, mass conservation bounds how large that boost can be.

6. Protocell dynamics

6.1 Growth and division

A condensate droplet that sustains net internal replication grows: material produced inside increases its volume. A *passive* liquid droplet, however, does not divide on its own — surface tension drives it to minimise area, so it stays spherical or coalesces with its neighbours. Genuine membrane-free division requires a **non-equilibrium drive**: continuous chemical turnover that destabilises large droplets (Zwicker et al. 2017), or an external thermal gradient that pinches a droplet in two (§9.4; Ianeselli et al. 2022). In the toy protocell model of this section we therefore *impose* division as a simple size threshold — a stand-in for those mechanisms — and ask what population dynamics follow.

```

In[*]:=
figProtoGrowthDivision[]

Out[*]=
Graphics-

```

Figure 6.1 Schematic of the protocell life cycle. Small condensates grow by accretion of material from the bulk and by internal replication. When they exceed a critical size, an imposed threshold — standing in for the chemically- or

thermally-driven fission of S9 — splits them into two daughters. Daughters inherit (with variation) the parent's composition, including its partition coefficient K and replication efficiency. Weaker daughters dissolve; fitter ones survive and grow.

6.2 Proto-Darwinian selection

If daughter droplets inherit their parent's chemical composition (approximately), and if that composition affects their fitness (replication rate, stability), then we have the three ingredients of natural selection: **variation** (daughters differ slightly from each other), **inheritance** (daughters resemble parents more than random droplets), and **differential fitness** (some compositions grow faster than others). This is **proto-Darwinian selection** — natural selection operating on condensate populations before the invention of genetic material.

We simulate this by initialising a population of 50 protocells with random partition coefficients K and replication rates, then letting them grow, divide, compete for resources, and either survive or dissolve:

```
In[ ]:=
figProtoSelection[]

Out[ ]=
-Graphics-
```

Figure 6.2 Proto-Darwinian selection in a simulated protocell population. Left: mean partition coefficient K increases over time as protocells with higher K out-compete those with lower K. Centre: mean replication rate increases. Right: lineage depth (max generation number) grows as successful lineages proliferate. These trends are the signature of adaptive evolution operating on membrane-free liquid droplets.

7. Condensates and disease

7.1 Liquid to gel to aggregate

Functional condensates are **liquid**: they flow, they exchange material rapidly with the surrounding cytoplasm, and they dissolve when the cell no longer needs them. But this liquidity is not guaranteed. Over time, or in the presence of disease-associated mutations, condensates can undergo a **maturation** transition: liquid → gel → solid aggregate. The gel state has reduced dynamics; the solid aggregate is essentially irreversible.

This maturation is implicated in several **neurodegenerative diseases**, though the pathologies differ. FUS and TDP-43 form functional liquid condensates whose mutated or mislocalised forms harden into the cytoplasmic **inclusions** of ALS and frontotemporal dementia (FTLD); tau matures into the neurofibrillary **tangles** of Alzheimer's and other tauopathies. (The amyloid **plaques** of Alzheimer's, by contrast, are built from a different peptide, amyloid-β.) In each case disease mutations or conditions shift the phase boundaries, accelerating the liquid→gel→solid transition toward these pathological end states.

We model this as a three-state kinetic system:



```
In[*]:=
figDiseaseTransition[]

Out[*]=
Labeled[Graphics, Condensate ageing: liquid ï¿½ gel ï¿½ aggregate\
> (neurodegeneration), Top]
```

Figure 7.1 Three kinetic scenarios for condensate ageing. Left: wild-type protein with slow maturation (healthy neuron). Centre: risk variant with moderate rates. Right: disease mutant (FUS/TDP-43 in ALS) with fast maturation — the liquid phase is rapidly depleted, and toxic aggregates accumulate.

7.2 Phase diagram of material states

The condensate material state depends on protein concentration and temperature (or interaction strength). We construct a schematic phase diagram showing the four regimes: mixed (one phase), liquid condensate, gel, and solid aggregate. Disease mutations effectively shift the boundaries, expanding the gel and aggregate regions at the expense of the functional liquid state.

```
In[*]:=
figDiseasePhaseDiagram[]

Out[*]=
Legended[Graphics, Placed[LineLegend[{Directive[PointSize[ $\frac{1}{90}$ ],
> AbsoluteThickness[2], Thickness[Large], RGBColor[0, 0, 1]],
> Directive[PointSize[ $\frac{1}{90}$ ], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1, 0.5, 0]], Directive[PointSize[ $\frac{1}{90}$ ], AbsoluteThickness[2],
> Thickness[Large], RGBColor[1, 0, 0]]},
> {Liquid condensate, Gel transition, Solid aggregate},
> LegendMarkers ->
> {{False, Automatic}, {False, Automatic}, {False, Automatic}},
> Joined -> {True, True, True}, LabelStyle -> {}, LegendLayout -> Column]\
> , After, Identity]]
```

Figure 7.2 Condensate material state phase diagram. Higher concentration and lower temperature (stronger interactions) drive the progression from mixed solution to liquid condensate to gel to solid aggregate. Disease mutations shift the boundaries toward lower concentrations and higher temperatures, making pathological aggregation more likely.

8. Nonlinear dynamics: stability, bifurcation, and coarsening

The previous sections built the model; here we analyse it with the toolkit of **nonlinear dynamics** — linear stability, bifurcation theory, Lyapunov functionals, and scaling laws. Every figure in this section is generated by `wolfram/nonlinear_dynamics.wls` (with a parallel Python implementation in `src/nonlinear_analysis.py`).

8.1 Linear stability and the dispersion relation

Linearise the Cahn–Hilliard equation about a uniform state $\phi = \phi_0 + \delta \exp(ik \cdot x + \omega t)$. Each Fourier mode grows or decays independently at the rate

$$\omega(k) = -M k^2 (f''(\phi_0) + \kappa k^2)$$

A uniform state is **unstable** precisely when $f''(\phi_0) < 0$ — the **spinodal** condition — in which case modes with $0 < k < k_c = \sqrt{-f''(\phi_0)/\kappa}$ grow. The fastest-growing mode and its rate are $k^* = \sqrt{-f''(\phi_0)/2\kappa}$ and $\omega^* = M (f''(\phi_0))^2/4\kappa$, setting the initial droplet spacing $\lambda^* = 2\pi/k^*$. This single mode is the seed of all condensate morphology.

The dispersion relation, live — the positive lobe (and hence the fastest-growing mode) appears exactly when $f'' < 0$:

In[]:=

```
 $\omega[k_, fpp_, M_, \kappa_] := -M k^2 (fpp + \kappa k^2);$ 
Plot[Evaluate@Table[ $\omega[k, fpp, 1, 1]$ , {fpp, {-2., -1., -0.5, 0.5}}],
  {k, 0, 1.6}, Frame -> True, FrameLabel -> {"k", " $\omega(k)$ "},
  PlotLegends -> {"f'=-2", "f'=-1", "f'=-0.5", "f'=+0.5 (stable)"}]
```

Out[]:=

```
Legended[-Graphics-, Placed[LineLegend[{Directive[Opacity[1.],
>   RGBColor[0.24, 0.6, 0.8], AbsoluteThickness[2]],
>   Directive[Opacity[1.], RGBColor[0.95, 0.627, 0.1425],
>   AbsoluteThickness[2]], Directive[Opacity[1.],
>   RGBColor[0.455, 0.7, 0.21], AbsoluteThickness[2]],
>   Directive[Opacity[1.], RGBColor[0.922526, 0.385626, 0.209179],
>   AbsoluteThickness[2]]},
> {f'=-2, f'=-1, f'=-0.5, f'=+0.5 (stable)}, LegendMarkers -> None,
> LabelStyle -> {}, LegendLayout -> Column], After, Identity]]
```

```

In[*]:=
figDispersion[]

Out[*]=
Legended[Graphics, Placed[LineLegend[{Directive[Opacity[1.],
> AbsoluteThickness[2], Thickness[Large], RGBColor[0., 0., 0.]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[0.455892, 0.160603, 0.459358]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[0.864723, 0.336476, 0.18292]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[0.991751, 0.5902, 0.0983158]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1., 0.842326, 0.418829]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1., 1., 1.]]},
> {I_0 = 0.06, I_0 = 0.1, I_0 = 0.2, I_0 = 0.3, I_0 = 0.45, I_0 = 0.6},
> LegendMarkers -> None, LabelStyle -> {}, LegendLayout -> Column],
> {0.18, 0.3}, Identity]]
    
```

Figure 8.1 Cahn–Hilliard dispersion relation for the Flory–Huggins free energy ($\chi = 1.5$). Curves with a positive lobe are unstable; dots mark the fastest-growing mode k^* . As the mean composition ϕ_0 approaches the spinodal edge, the growth rate collapses to zero and the selected wavelength diverges.

8.2 Mode selection across the parameter plane

The selected wavelength $\lambda^* = 2\pi \sqrt{2\kappa/(-f'')}$ grows with the surface-tension parameter κ and diverges at the spinodal boundary, where $f'' \rightarrow 0$. The map below shows how the length scale is selected jointly by composition and κ .

```

In[*]:=
figModeSelection[]

Out[*]=
Legended[Graphics, Placed[BarLegend[{Blend[SunsetColors, #1] & , {3/2, 25}}},
> LabelStyle -> {}, LegendLayout -> Column, LegendMarkerSize -> 620,
> ScalingFunctions -> {Identity, Identity}, Charting`AxisLabel -> None,
> Charting`TickSide -> Right, ColorFunctionScaling -> True], After,
> Identity]]
    
```

Figure 8.2 Fastest-growing wavelength λ^* over the (ϕ_0, κ) plane (clipped at 25 for contrast). The white curve is the spinodal $f'' = 0$; outside it the homogeneous state is linearly stable and no length scale is selected.

8.3 Bifurcation: common-tangent coexistence vs. the spinodal

Sweeping the control parameter traces out the **bifurcation diagram**. For the symmetric double well $f = -a/2 \phi^2 + b/4 \phi^4$ the coexistence amplitude follows a supercritical pitchfork: the binodal $\phi = \pm \sqrt{a/b}$ encloses the spinodal $\phi = \pm \sqrt{a/3b}$. For Flory–Huggins the coexistence curve is the **common-tangent** (Maxwell) construction. Because the osmotic pressure is linear in χ , the dense

branch follows the closed form $\chi(\phi_R) = (\phi_R g(\phi_R) - h(\phi_R))/\phi_R^2$, which is numerically robust where the repository's earlier two-variable solver failed.

```
In[*]:=
figNLPhaseDiagram[]

Out[*]=
-Graphics-
```

Figure 8.3 (a) Flory–Huggins phase diagram: binodal (coexistence, blue) outside the spinodal (instability, red dashed), meeting at the critical point; the orange dot marks the solver operating point ($\phi = 0.3$, $\chi = 1.5$). (b) The symmetric double-well pitchfork, with the metastable region between binodal and spinodal.

8.4 The free energy as a Lyapunov functional

Cahn–Hilliard dynamics are a **gradient flow** of the free-energy functional $F[\phi] = \int [f(\phi) + (\kappa/2) |\nabla \phi|^2] dx$. Conservation of ϕ and the structure of the equation guarantee $dF/dt \leq 0$: F is a **Lyapunov functional** and the phase-separated state is its minimiser. Each coarsening event — two domains merging — appears as a step down in F .

```
In[*]:=
figEnergyLyapunov[]

Out[*]=
-Graphics-
```

Figure 8.4 (a) Space–time evolution of the 1D double-well Cahn–Hilliard equation (NDSolve): domains form, then coarsen by merging. (b) The free energy $F[\phi]$ decreases monotonically, falling in discrete steps as domains coalesce — the signature of a Lyapunov functional. (Note: the strongly stabilised Flory–Huggins solver instead freezes; the canonical $t^{1/3}$ coarsening, with $L^3 = L_0^3 + K t$ linear in time, is reproduced by the double-well Model B in the Python implementation.)

8.5 Mass-conserving reaction enhancement

Section 4 estimated the bimolecular rate boost inside a condensate as $K_A K_B$. That is the *open-system* limit (buffered reservoir). Imposing **mass conservation** — the condensate concentrates reactants but *depletes* the surrounding dilute phase — gives a very different answer. With dense area fraction p and rapid partitioning $c_{in} = K c_{out}$ in each phase, the volume-averaged rate relative to a well-mixed system is

$$\frac{k_{\text{eff}}}{k_{\text{out}}} = \frac{p K_A K_B + (1-p)}{(1+p(K_A-1))(1+p(K_B-1))}$$

The enhancement vanishes at both $p \rightarrow 0$ (no condensate) and $p \rightarrow 1$ (everything uniform); it is maximal at an **intermediate dense fraction** $p^* \approx 1/K$, where the gain is only $\approx K/4$ — not the naive K^2 . Bulk depletion is the hard ceiling on condensate catalysis, and it predicts an optimal droplet volume fraction — a falsifiable, quantitative claim.

```
In[*]:=
figReactionEnhancement[]

Out[*]=
-Graphics-
```


Figure 8.5 Mass-conserving rate enhancement $k_{\text{eff}}/k_{\text{out}}$. (a) Dependence on the partition coefficients at fixed dense fraction $p = 0.05$. (b) Dependence on dense fraction for several K : each curve peaks at $p^* \approx 1/K$ with a maximum gain of order $K/4$, far below the open-system K^2 .

9. Active and chemically-maintained condensates

Living condensates are not at equilibrium: ATP-driven reactions continually build up and break down their components. This section extends the model with chemistry. Code: `src/active_models.py`, `src/fh_extensions.py`, and `wolfram/active_dynamics.wls`.

9.1 Reaction-arrested coarsening

Add a linear chemical reaction that interconverts droplet material and a soluble form, $s(\phi) = k_+(1-\phi) - k_-\phi = -k_{\text{react}}(\phi - \phi_{\text{ss}})$, to the conserved dynamics:

$$\frac{\partial \phi}{\partial t} = M \nabla^2 \mu - k_{\text{react}} (\phi - \phi_{\text{ss}})$$

The reaction adds a uniform $-k_{\text{react}}$ to the growth rate, $\omega(k) = -Mk^2(f_{\text{H}} + \kappa k^2) - k_{\text{react}}$, which **stabilises the long-wavelength modes**: the unstable band becomes finite and a length scale is selected. Ostwald ripening is arrested and the system settles into a steady-state emulsion — the mechanism behind chemically-active droplets (Zwicker et al. 2015; Bauermann et al. 2025).

```
In[ ]:=
figActiveDispersion[]

Out[ ]:=
Legended[Graphics, Placed[LineLegend[{Directive[Opacity[1.],
> AbsoluteThickness[2], Thickness[Large], RGBColor[0., 0., 0.]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[0.788287, 0.259816, 0.270778]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1., 0.682688, 0.129771]],
> Directive[Opacity[1.], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1., 1., 1.]]},
> {k_react = 0., k_react = 0.05, k_react = 0.15, k_react = 0.25},
> LegendMarkers -> None, LabelStyle -> {}, LegendLayout -> Column],
> {0.25, 0.28}, Identity]]
```

Figure 9.1 Reaction-modified dispersion relation. Increasing the reaction rate pushes the small- k (long-wavelength) modes below zero, leaving a finite unstable band — coarsening can no longer proceed to arbitrarily large domains.

```
In[ ]:=
figReaction1D[]

Out[ ]:=
Graphics-
```

Figure 9.2 1D reaction–Cahn–Hilliard (NDSolve). (a) Without reaction the domains coarsen toward a single domain. (b) With reaction the pattern freezes into a steady-state periodic structure of selected wavelength.

The full 2D problem (solved with a spectral method in `src/active_models.py`) sharpens the same picture: chemical turnover turns the coarse, ever-growing morphology into a fine **steady-state emulsion** whose mean droplet size shrinks with reaction rate, and whose size distribution is narrower than the passive LSW one (Bauermann et al. 2025).

9.2 Active Model B: broken detailed balance

A genuinely non-equilibrium term $\lambda|\nabla\phi|^2$ added to the chemical potential breaks detailed balance (Active Model B; Tjhung et al. 2018). Measured honestly, this term shifts the coexistence densities **asymmetrically** (the dense phase becomes denser than the dilute phase is dilute) and modestly slows coarsening, but does *not* by itself arrest it — the robust route to a steady-state emulsion is the chemical reaction of §9.1.

9.3 Stickers and spacers: multivalency drives phase separation

Real biomolecules interact through specific, saturable contacts ("stickers" joined by flexible "spacers"; Harmon & Pappu 2017). Adding a saturable association free energy $f_{\text{bond}} = \phi_s [\ln(1-p) + p/2]$ (bond fraction p from the law of mass action $p/(1-p)^2 = K_a \phi_s, \phi_s = v\phi$) to the Flory–Huggins free energy expands the two-phase region and lowers the critical interaction: higher sticker valence v makes phase separation far easier.

`In[*]:=`

`figStickers[]`

`Out[*]=`

```

Legended[Graphics, Placed[LineLegend[{Directive[PointSize[ $\frac{1}{120}$ ],
> AbsoluteThickness[2], Thickness[Large], RGBColor[0., 0., 0.]],
> Directive[PointSize[ $\frac{1}{120}$ ], AbsoluteThickness[2], Thickness[Large],
> RGBColor[0.788287, 0.259816, 0.270778]],
> Directive[PointSize[ $\frac{1}{120}$ ], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1., 0.682688, 0.129771]],
> Directive[PointSize[ $\frac{1}{120}$ ], AbsoluteThickness[2], Thickness[Large],
> RGBColor[1., 1., 1.]]},
> {no stickers (FH), v=2, Ka=2, v=3, Ka=2, v=4, Ka=4},
> LegendMarkers ->
> {{False, Automatic}, {False, Automatic}, {False, Automatic},
> {False, Automatic}}, Joined -> {True, True, True, True},
> LabelStyle -> {}, LegendLayout -> Column, {0.78, 0.78}, Identity]]

```

Figure 9.3 Stickers-and-spacers spinodal curves. Each curve bounds the two-phase region; increasing sticker valence pushes the critical point down and outward — multivalency drives liquid–liquid phase separation even when simple mixing is unfavourable.

9.4 Thermal-gradient fission: membrane-free division

A temperature gradient makes the interaction $\chi \sim 1/T$ position-dependent. A warm band (weakened cohesion) drawn through an elongated droplet dissolves its neck and splits it into two daughters — division without a membrane, a candidate protocell mechanism (laneselli 2022). The split is genuinely band-driven: two domains form with the band, only one without it.

```
In[*]:=
figThermalFission[]

Out[*]=
-Graphics-
```

Figure 9.4 Thermal-gradient fission (1D Cahn–Hilliard, NDSolve). (a) At uniform temperature a single domain persists. (b) A central warm band, where the well depth $a(x)$ dips below zero, dissolves the neck and splits the domain into two daughters — membrane-free division.

9.5 The dispersion surface

Finally, the linear-stability result of §8 viewed as a surface: the growth rate $\omega(k, \phi_0)$ over the whole plane of wavenumber and composition, with the neutral-stability ridge $\omega=0$ marking the boundary of the unstable region.

```
In[*]:=
figDispersionSurface[]

Out[*]=
-Graphics3D-
```

Figure 9.5 The Cahn–Hilliard dispersion surface $\omega(k, \phi_0)$ for the Flory–Huggins free energy. The unstable dome shrinks to nothing as the composition approaches the spinodal edge; the cyan curve is the neutral-stability ridge.

10. Emergent complexity: selection, patterns and cycles

The final layer of the model lets genuinely life-like behaviour emerge from the same physics. Code: `src/multicomponent.py`, `src/darwinian_selection.py`, `src/reaction_diffusion_cargo.py`, `src/wet_dry_cycling.py`, `src/parameter_space.py`, and the Wolfram companion `wolfram/phase3_extensions.wls`.

10.1 Multi-component condensates

Real condensates contain dozens of macromolecules. For N mutually attracting components that co-condense in near-stoichiometric proportions, the demixing entropy is reduced as if they were one effective polymer of length N , and the symmetric Flory–Huggins critical point migrates (Qian & Knowles 2022):

$$\phi_c = \frac{1}{1 + \sqrt{N}}, \chi_c = \frac{2(1 + \sqrt{N})^2}{N}$$

As N grows the critical concentration falls toward zero — multivalent mixtures phase-separate at vanishing concentration — and the critical interaction drops, so they demix far more easily. Distinct scaffolds give distinct condensates, and each client partitions selectively into the one it binds best (the physical basis of organelle identity).

```
In[*]:=
figMulticomponent[]

Out[*]=
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> AbsoluteThickness[2], RGBColor[0.268411, 0.097776, 0.364682],
> Thickness[Large]], Directive[Opacity[1.], AbsoluteThickness[2],
> RGBColor[0.771667, 0.254855, 0.280207], Thickness[Large]],
> Directive[Opacity[1.], AbsoluteThickness[2],
> RGBColor[0.983502, 0.497711, 0.0668605], Thickness[Large]],
> Directive[Opacity[1.], AbsoluteThickness[2],
> RGBColor[1., 0.770489, 0.288753], Thickness[Large]]}],
> {N=2, N=3, N=5, N=10}, LegendMarkers -> None, LabelStyle -> {},
> LegendLayout -> Column], {0.82, 0.7}, Identity]]
```

Figure 10.1 Multi-component spinodals for $N=2,3,5,10$ in the Qian-Knowles convention; the critical point (white dots) migrates as $\phi_c = 1/(1 + \sqrt{N})$ toward low concentration and low interaction as the number of components grows.

```
In[*]:=
figCompartmentalization[]

Out[*]=
Graphics-
```

Figure 10.2 Selective compartmentalisation: a ternary (A, B, solvent) free energy in which two mutually-repelling solutes form an A-rich and a B-rich condensate — each client partitions into the droplet it binds best, the physical basis of organelle identity.

10.2 Darwinian selection among protocells

The intellectual climax. Replicators partition into condensate droplets and replicate at a rate set by a heritable genotype g , competing for a **shared global nutrient** under a uniform chemostat dilution. Replicators whose growth rate $r_0 g$ falls below the dilution rate are out-competed (competitive exclusion), and the population's mean fitness climbs. A full 2D spatial version (`src/darwinian_selection.py`), in which fuel-driven droplets grow, divide under periodic thermal pulses, and pass their genotype to daughters, reaches the same outcome with explicit lineages.

The result is natural selection from pure physics. The cargo-weighted mean fitness rises over generations and saturates at a mutation-selection balance — no biology is hand-coded; selection is an emergent consequence of the dynamics.

What the selection requires. The effect is not automatic: it demands **Malthusian (per-capita) replication competing for a shared, global nutrient** together with a uniform chemostat dilution. Replacing the shared resource with a *local* carrying capacity, or raising the mutation rate too high, erases the inheritance signal — mean fitness then drifts rather than climbs and selection collapses. The trajectory below is therefore a genuine result of the chosen ecology, not a foregone conclusion.

```
In[*]:=
figSelectionChemostat[]

Out[*]=
-Graphics-
```

Figure 10.3 Chemostat competition (*NDSolve*). Left: the population's mean fitness $\langle g \rangle$ rises toward g_{max} as fitter replicators take over. Right: the fittest genotype out-competes the rest — competitive exclusion, with no biology hand-coded.

```
In[*]:=
figLineage[]

Out[*]=
Graph[<14>, <13>]
```

Figure 10.4 A protocell phylogeny generated by the same selection dynamics: fit (yellow) lineages branch and proliferate while low-fitness (blue) lineages die out (node colour = fitness).

10.3 Reaction-diffusion inside condensates

A condensate is also a chemical reactor. When crowding slows the activator far more than the inhibitor, the diffusion ratio D_v/D_u crosses the **Turing threshold** *only inside* the dense phase: a band of wavenumbers becomes unstable there while every mode stays stable in the dilute surroundings, so spots self-organise inside the condensate and nowhere else. (The full 2D patterns, and the reaction hotspot that forms in a shell at the droplet interface, are computed in `src/reaction_diffusion_cargo.py`.)

```
In[*]:=
figTuring[]

Out[*]=
Legended[-Graphics-, Placed[LineLegend[{Directive[Opacity[1.],
> AbsoluteThickness[2], RGBColor[0.78, 0.12, 0.36], Thickness[0.01]],
> Directive[Opacity[1.], AbsoluteThickness[2],
> RGBColor[0.12, 0.44, 0.7], Thickness[0.008], Dashing[{Small, Small}]]],
> , {inside droplet (dense): unstable band,
> dilute phase: all modes stable}, LegendMarkers -> None,
> LabelStyle -> {}, LegendLayout -> Column], {0.62, 0.25}, Identity]]
```

Figure 10.5 Schnakenberg linear-stability growth rate $\lambda(k)$: inside the dense phase a band of wavenumbers is unstable (patterns form), while in the dilute phase every mode is stable. The Turing instability is switched on by the condensate.

10.4 Wet-dry cycling

On the early Earth, rock pools cycled between flooded (wet) and evaporated (dry) states. Cycling the interaction $\chi(t) = \chi_0 + \Delta\chi \sin(2\pi t/T)$ swings the system between a one-phase well (dissolve, remix) and a spinodally unstable double well (re-nucleate many droplets). Each dissolution erases the size differences that drive Ostwald ripening, so a **population** of droplets is maintained, while a matched constant- χ run coarsens toward a single winner (laneselli 2022).

```

In[*]:=
figWetDry[]

Out[*]=
Legended[-Graphics-, Placed[BarLegend[{Blend[RedBlueTones, #1] & ,
>      {- (69/50), 69/50}], LabelStyle -> {}, LegendLayout -> Column,
>      LegendMarkerSize -> 408., ScalingFunctions -> {Identity, Identity},
>      Charting`AxisLabel -> None, Charting`TickSide -> Right,
>      ColorFunctionScaling -> True], After, Identity]]

```

Figure 10.6 1D wet-dry Cahn-Hilliard space-time (NDSolve): with a cyclic well depth $a(t)$, domains dissolve at each wet half-cycle and re-nucleate at each dry half-cycle. The repeated dissolution resets the size differences that would otherwise drive Ostwald ripening, so a population of droplets is sustained rather than coarsening to a single winner (laneselli et al. 2022).

10.5 The accessible parameter space

Finally, two analytic maps of the model's regimes: the (χ, N) plane separating one-phase from two-phase by the critical line $\chi_c(N)$, and the (K, p) mass-conserving rate-enhancement landscape with its optimal ridge $p^* \approx 1/K$. (The full four-panel regime atlas — which also includes the thermodynamic (ϕ, θ, κ) plane and the active dissolution / microphase / macrophase plane from 2D simulations — is built in `src/parameter_space.py`.)

```

In[*]:=
figParamSpace[]

Out[*]=
-Graphics-

```

Figure 10.7 The two analytic regime planes. Left, (χ, N) : the one-phase / two-phase boundary along the critical line. Right, (K, p) : the mass-conserving rate enhancement, with the optimal ridge $p^* \approx 1/K$ (dashed).

11. Conclusions

What we showed. (1) The Flory-Huggins lattice theory, implemented in a few lines of Wolfram Language, correctly predicts the qualitative phase behaviour of biomolecular condensates: a UCST phase diagram with the critical concentration shifting to lower values for longer/more multivalent molecules. (2) The Cahn-Hilliard equation, solved natively by NDSolve, captures spinodal decomposition; the subsequent Lifshitz-Slyozov $t^{1/3}$ coarsening is reproduced with the double-well Model B form, since the stiff Flory-Huggins solver freezes near the pure phases. (3) The concentration-enhancement argument gives a K^2 boost for bimolecular reactions in the open-system limit — a partial resolution of the dilute-ocean problem, since mass conservation caps the realised gain at order $K/4$ (§8.5). (4) Protocell simulations show that condensate populations can undergo proto-Darwinian selection without genetic material. (5) The liquid-to-gel-to-aggregate transition connects condensate physics to neurodegenerative disease. (6) Layering the model — multi-component mixtures, a replicating cargo with heredity and a shared nutrient, reaction-diffusion, and wet-dry cycling — makes genuinely life-like behaviour emerge: **natural selection, Turing morphogenesis, and ripening-resistant droplet populations all fall out of the same physics.**

The bigger picture. Biomolecular condensates sit at the intersection of soft-matter physics, molecular biology, the origin of life, and neurodegeneration. The mathematical framework — Flory-

Huggins thermodynamics, Cahn–Hilliard dynamics, Michaelis–Menten kinetics, Lifshitz–Slyozov coarsening — is classical and well-understood. What is new is the recognition that this physics operates inside living cells, that it provides the thermodynamic foundation for protocellular compartments, and that its failure modes are disease. The condensate revolution is not a discovery of new physics; it is the discovery that old physics was hiding in plain sight inside biology.

Key takeaways.

- **Phase separation is thermodynamics, not biology.** A two-parameter Flory–Huggins free energy already predicts when proteins and RNA demix into condensates.
- **Concentration is the payoff — but it is bounded.** Partitioning boosts bimolecular rates by up to K^2 in the open-system limit, while mass conservation caps the realised gain at order $K/4$.
- **Life-like behaviour is emergent.** Add chemistry, heredity and a shared resource and natural selection, Turing patterns and ripening-resistant populations fall out of the same equations — no biology hand-coded.
- **The same physics, run backwards, is disease.** The liquid→gel→solid maturation that endangers neurons is the flip side of the transition that organises healthy cells.

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Further reading

For an accessible, non-technical tour of the same story – condensates, coacervates and the idea that the origin of life may have been hiding inside our own cells all along – see the *New Scientist* cover feature that prompted this notebook:

Marshall, M. (2026). Hiding in plain sight. *New Scientist*, 20 June 2026, pp. 30–33 (Features / Cover story). Online edition titled “Has the answer to life’s origins been hiding in our cells all along?” [newscientist.com/article/2529162](https://www.newscientist.com/article/2529162) (<https://www.newscientist.com/article/2529162-has-the-answer-to-lifes-origins-been-hiding-in-our-cells-all-along/>).

13. Reproducibility

All source code lives in the repository github.com/mthiel74/biomolecular-condensates (<https://github.com/mthiel74/biomolecular-condensates>). The pipeline has two parallel tracks: Wolfram

Language (analysis scripts in `wolfram/` and this notebook builder in `community/`) and Python (`src/`, including a full 2D Cahn–Hilliard spectral solver). The figures come from both tracks: the Wolfram scripts write directly to `docs/images/`, while the Python scripts write to `figures/` and are copied across. Wolfram Language 13+ is required. To regenerate this notebook from scratch, run the Python track first, then the Wolfram runner (which now also executes the Phase-3 companion and copies the Python figures into place), then the builder:

```
# 1. Python track → writes figures to figures/
pip install -r src/requirements.txt # numpy, scipy, matplotlib (+ ffmpeg for
animations)
python3 src/flory_huggins.py
python3 src/cahn_hilliard.py
python3 src/condensate_kinetics.py
python3 src/protocell.py
python3 src/nonlinear_analysis.py # nonlinear-dynamics analysis ($8)
python3 src/fh_extensions.py # stickers-and-spacers ($9.3)
python3 src/active_models.py # active condensates ($9)
python3 src/multicomponent.py # multi-component condensates ($10.1)
python3 src/darwinian_selection.py # Darwinian selection ($10.2)
python3 src/reaction_diffusion_cargo.py # reaction-diffusion ($10.3)
python3 src/wet_dry_cycling.py # wet-dry cycling ($10.4)
python3 src/parameter_space.py # parameter space ($10.5)
python3 src/phase3_heroes.py # hero visualisations

# 2. Wolfram analysis + copy of Python figures → docs/images/
wolframscript -file wolfram/run_all.wls

# 3. Rebuild this notebook
wolframscript -file community/build_notebook.wls
```

Licence and provenance. All code and every figure in this notebook are original and generated by the accompanying scripts; the notebook and figures are released under the MIT licence (see LICENSE in the repository). Suggested Wolfram Community tags: *Biology, Physics, Differential Equations, NDSolve, Phase Transitions, Origin of Life, Soft Matter*.

Generated on June 26, 2026 with Wolfram Language 15..